

EVALUATION OF SPAWNING SUBSTRATE ENHANCEMENT FOR WHITE STURGEON
IN A REGULATED RIVER: EFFECTS ON LARVAL RETENTION AND DISPERSALJ. A. CROSSMAN^{a*} AND L. R. HILDEBRAND^b^a BC Hydro, Water License Requirements, Castlegar, BC, Canada^b Golder Associates Ltd, Castlegar, BC, Canada

ABSTRACT

Site-specific habitat alterations have improved spawning success and early life stage survival of different fish species, including sturgeon, in regulated rivers. We modified the substrate within a section of river at the only known spawning site used by white sturgeon (*Acipenser transmontanus*) in the Mid Columbia River, Canada. Existing armoured riverbed conditions were modified using a mixture of larger and smaller angular rock with the assumption that the larger material would remain in place at higher discharges and help retain the smaller material. This increased substrate complexity and the amount of available interstitial spaces. We stocked 2-day posthatch larvae over both the modified site and at an adjacent control site that represented existing substrate conditions. Our objectives were to determine (i) the extent that stocked larvae remained in both the modified and control sites immediately after release, (ii) the timing of subsequent dispersal of larvae from both sites and (iii) how total length of dispersing larvae changed over time and by site. Results from this work indicated that the modified section of riverbed retained significantly higher numbers of larvae after release compared with the control site. Larvae at the modified site were able to hide and remain within the substrate and initiated downstream drift 15 days after release. With the exception of the first day after release, dispersal from both sites occurred at night. There was a significant effect of time after release and site on the total length of dispersing larvae. The larger variation in total larval length observed at the control site compared with the modified site indicated greater difficulty in hiding within the control substrate. Larvae initiated dispersal from the modified site at a mean size of 17.5 mm, which may indicate an important growth threshold before drift. Results from this work are important for future mitigative efforts for sturgeon in regulated rivers where changes to spawning substrates have occurred. Copyright © 2012 John Wiley & Sons, Ltd.

KEY WORDS: river restoration; *Acipenser transmontanus*; larval stocking; larval growth; bed armouring; flow regulation

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INTRODUCTION

The regulation of rivers has altered the ecosystem processes important for the persistence of many aquatic species (Bain *et al.*, 1988; Ligon *et al.*, 1995; Poff *et al.*, 1997; Osmundson *et al.*, 2002). Physical habitat in rivers is critically important to ecosystem function and is more sensitive to change compared with other ecosystems (Ligon *et al.*, 1995). The regulation of rivers typically results in an increase in substrate armouring and embeddedness. The armouring of channel beds below dams typically occurs within the first 5–10 years after dam closure (Williams and Wolman 1984 as cited in Juracek, 2001). As substrates become more armoured and embedded, the interstitial space between substrate particles is reduced, which alters channel bedform and hydraulics and reduces streambed roughness (Sylte and Fischenich, 2002). Without periodic mobilisation of fine sediments from the coarse bed material, which would have occurred during preregulation freshet or flood events, deposited fines eventually clog interstitial voids (Osmundson *et al.*,

2002). Decreasing interstitial space limits the available area and cover for fish eggs and larvae of multiple species, affecting embryo survival, larval emergence and physical size (Tappel and Bjornn, 1983; Young *et al.*, 1990; Lisle and Lewis, 1992; Merz and Setka, 2004). Results from studies conducted to date indicate that restoration activities focussed on improving physical habitat conditions in regulated rivers is warranted and critical to the persistence of many fish species.

The establishment of a natural flow regime may provide the largest single benefit to habitat restoration in many river systems (Poff *et al.*, 1997). However, for many regulated rivers, this is not possible because of issues around flood protection and power generation. In these situations, restoration or enhancement projects typically focus on structural improvements (Fuselier and Edds, 1995; Gore *et al.*, 1998), gravel augmentation (Wheaton *et al.*, 2004; Merz and Ochikubo Chan, 2005) or other forms of physical habitat enhancement (Merz and Setka, 2004). Many enhancement projects have focussed on spawning habitat and have been successful in improving conditions necessary for spawning and early life stage survival (Dumont *et al.*, 2011; Roseman *et al.*, 2011). Furthermore, positive indirect effects on other ecosystem

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processes (e.g. macroinvertebrates; Merz and Ochikubo Chan, 2005) have been identified and demonstrate the importance of comprehensive designs when implementing habitat enhancement programs (Wheaton *et al.*, 2004). This is especially important for programs aimed at improving spawning habitat for imperiled species where reproductive success is poor.

White sturgeon (*Acipenser transmontanus*) are adapted to large river systems and are highly fecund broadcast spawners that typically spawn on the declining limb of the spring freshet (Hildebrand *et al.*, 1999; McAdam *et al.*, 2005) in turbulent areas with clean gravel, cobble or boulder substrates (Scott and Crossman, 1973; Parsley and Beckman, 1993; Hildebrand *et al.*, 1999). Several populations of white sturgeon are currently undergoing recruitment failure, including the Kootenai (Anders *et al.*, 2002), Nechako (McAdam *et al.*, 2005) and Upper Columbia Rivers (Irvine *et al.*, 2007). Although the specific causal factors responsible for recruitment failure are still poorly understood, recruitment failure is thought to occur at the early life stages (Gross *et al.*, 2002; Ireland *et al.*, 2002; McAdam *et al.*, 2005; Gregory and Long, 2008). The flow regulation and alteration of the natural hydrograph have resulted in numerous geomorphic changes to the riverine environment where white sturgeon spawn (Coutant, 2004; McAdam *et al.*, 2005; Paragamian *et al.*, 2005). Brannon *et al.* (1984) suggested that the infilling of substrate interstices may affect early life stage survival, and more recently, the potential linkage between changes to river bed substrates and survival of eggs and early larval stages has received increased attention as a factor limiting sturgeon recruitment in regulated rivers (McAdam *et al.*, 2005; McAdam, 2011).

White sturgeon residing in the Columbia River in Canada were federally listed as endangered in 2006 under the Species At Risk Act (2011). A small population segment of white sturgeon (50 adults; 95% CI = 37 – 92; Golder, 2006) resides in the Arrow Lakes Reservoir, located on the Columbia River between Hugh L. Keenleyside Dam and Revelstoke Dam. Adults spawn in the riverine section of the Columbia River downstream of Revelstoke Dam. Spawning does not occur annually, and natural recruitment to the larval stage has only been demonstrated twice in nine years of sampling (Golder, 2011). Following construction in 1974, the operation of Revelstoke Dam has resulted in substantial changes to downstream physical habitat. Current habitat conditions are considered unsuitable for egg incubation and early larval hiding because of a high degree of substrate armouring and embeddedness. Further changes to operations at Revelstoke Dam for additional power generation will increase peak daily river discharge and have the potential to result in additional changes to physical characteristics within the spawning, egg incubation and early larval rearing habitat. Given that this is the only documented location where this imperiled population segment spawns, investigations into spawning habitat enhancement are warranted.

To assess the suitability and effectiveness of habitat enhancement for white sturgeon, we stocked recently hatched larvae over both modified substrate and substrate that represented existing conditions. We focussed on the larval stage because of uncertainties in our ability to evaluate the modified habitats suitability for spawning given low numbers of adults and infrequent annual spawning. Specific objectives of this study were (i) to determine the extent that stocked larvae remained in both the control and modified sites immediately after release, (ii) to determine the timing of larval dispersal from both sites and (iii) to determine how rates of growth and yolk sac use changed over time after release.

STUDY SITE

Work was conducted in a section of the Mid Columbia River approximately 6 km downstream of Revelstoke Dam, immediately adjacent to the Revelstoke Golf Course, and just upstream from the Jordan River confluence (Figure 1). Flows in the Revelstoke spawning area are affected by two upstream dams. Mica Dam (completed in 1973) is a large water storage and power production facility that has essentially flattened the annual hydrograph and eliminated flood events. River hydraulics (water depth and velocity) over the study site are directly affected by discharge from Revelstoke Dam (completed in 1983), a daily peaking facility that releases water in response to hourly power demands. This facility typically operates from dawn to dusk, with daily discharges between 0 and 1700 m³/s. Surface water velocities at the study site can exceed 3 m/s during periods of peak generation (Golder, 2012).

In the white sturgeon spawning area, the riverbed has essentially undergone armouring since 1991, the highest flow event on record for Revelstoke Dam. This 4550 m³/s flow event resulted in a resuspension and redistribution of compacted riverbed substrates in the armoured channel, which produced large areas of clean loose gravels, cobbles and small boulders (RL&L, 1994). Subsequently, the riverbed has rearmoured to the maximum powerplant discharge of 1700 m³/s and become substantially more imbedded. The addition of a fifth generation unit at Revelstoke Dam in late 2011 and a sixth unit in the future will increase maximum river levels and flow velocities and will require the stabilisation of sections of river banks in downstream areas. One of these bank stabilisation areas was situated immediately adjacent to the white sturgeon spawning area. Bank protection at this location was completed by installing a riprap revetment along 1400 m of shoreline. The protection work was designed to withstand flows of 2500 m³/s, which represented the maximum discharge with six-units operational at Revelstoke Dam. The riprap project offered an opportunity to enhance substrate at the known spawning area and obtain information of the potential effects of flow regulation on spawning and

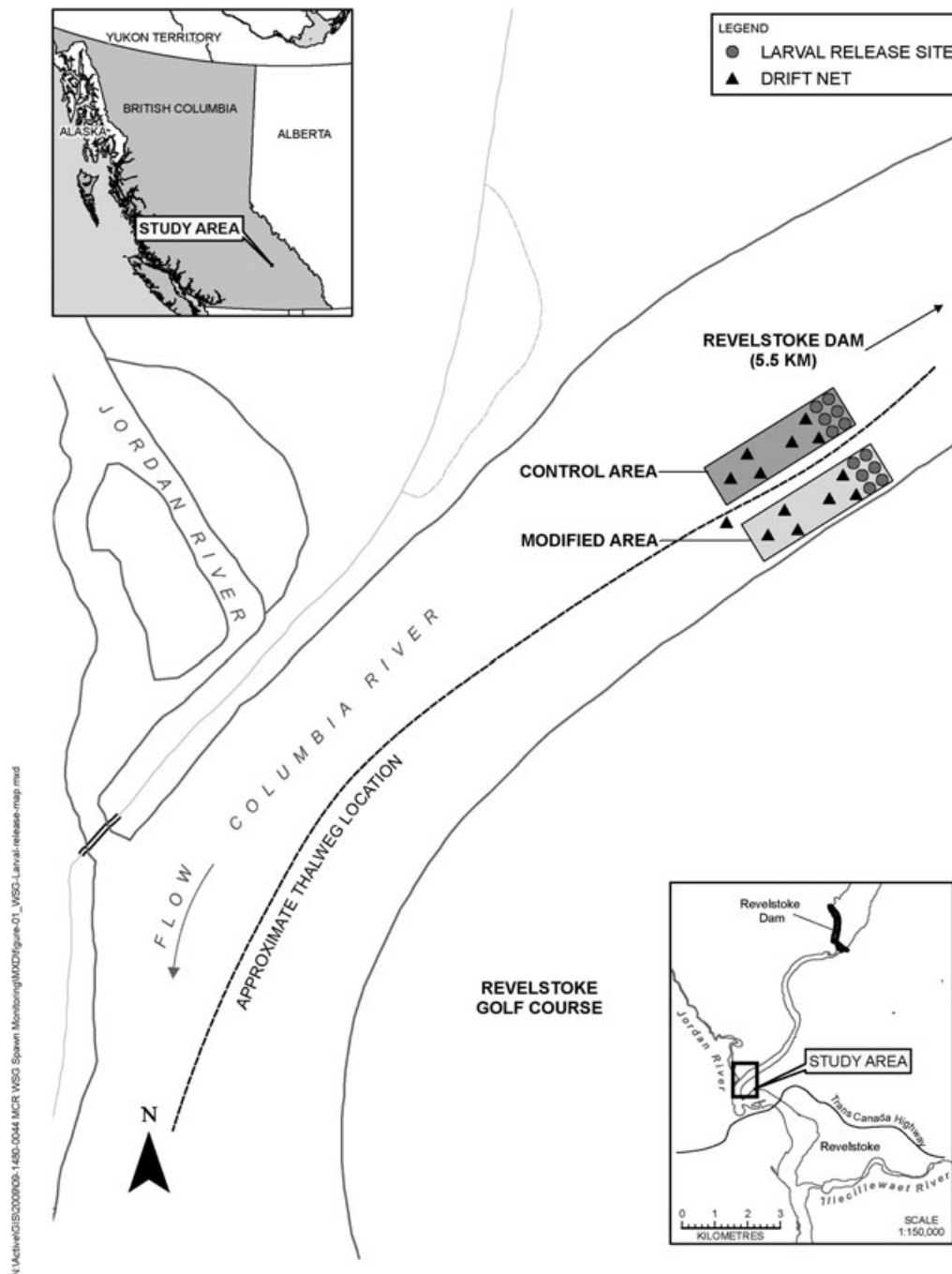


Figure 1. Map of the study site on the Columbia River near Revelstoke British Columbia showing the location of both the control and the modified sites and the drift net sampling design. Control and modified site dimensions measured 10 m in width by 100 m in length

early life stage incubation habitat. Efficiencies were available, both financially and logistically, given that crews and equipment were on site. Work was conducted simultaneously at two locations within the study site (Figure 1). These included the newly modified experimental site immediately adjacent to the Revelstoke Golf Course and a control site that represented existing habitat conditions.

METHODS

Spawning site modification

Site-specific habitat alterations have improved spawning success for other species of sturgeon (e.g. Lake sturgeon, *Acipenser fulvescens*; Folz and Meyers, 1985; Environnement Illimité Inc., 2010; Dumont *et al.*, 2011), where riprapping

was conducted to stabilise shoreline and for other fish species in regulated rivers (Roni *et al.*, 2002; Merz and Setka, 2004). We modified the substrate at the only known white sturgeon spawning site (Figure 1) in the Mid Columbia River. This site was chosen on the basis of 5 years of egg deposition data from this area (Golder, 2011). Specifics regarding spawning site modifications were developed using the best available data regarding white sturgeon spawning requirements and recommendations from a committee of technical experts. White sturgeon have been shown to spawn over substrates dominated by cobbles and boulders (Parsley *et al.*, 1993; Hildebrand *et al.*, 1999). Cobbles are generally defined as material 64–256 mm in diameter, whereas boulders are material greater than 256 mm in diameter (McMahon *et al.*, 1996). Discussions around the substrate modifications focussed on improving the complexity of the available habitat by increasing the amount of available interstitial spaces for use by early life stages of white sturgeon (i.e. eggs and larvae). This was accomplished by using larger boulders (covering 90% of the total area) that were expected to remain in place at higher discharges and help retain the smaller cobbles also added to the area. Further, the material used was angular in shape to provide more interstitial space when settled. The larger material was 200- to 300-mm diameter washed hard quarry stone that was free of cracks, seams or other structural defects with specific gravity not less than 2.60. The maximum dimension of any piece was not more than 2.5 times its least dimension. Smaller material consisted of blast rock (covering 10% of the total area) between 25 and 80 mm diameter. The rock used was tested using acid–base accounting and was found to be non-acid generating (Klohn Crippen Berger, 2009). The rock was placed in the river thalweg below the zero-discharge water level for the Columbia River at that location. This was required to avoid dewatering eggs or larvae, an identified impact of hydroelectric operations near this location (Golder, 2011) and others (Van der Leeuw *et al.*, 2006). Rock was placed in the river during hours of zero discharge using mini-excavators (Klohn Crippen Berger, 2009). The final modified site covered an area approximately 100 m in length by 10 m wide and was 0.6 m thick.

We compared the modified site with an adjacent control site that represented existing riverbed substrate conditions. The control site was parallel to, and placed 20 m offshore from, the modified site (Figure 1) to minimise probabilities of lateral movements of larvae between sites. The dimensions of the control site were identical to the modified site for the purposes of monitoring as described in the following sections. Substrate conditions were described using underwater videography transects completed over the control site, with a mean substrate diameter of 43.9 ± 3.12 mm (Golder, 2012). Substrate embeddedness at the control site was visually estimated at >75% based on underwater video footage.

Supplemental progeny rearing and transport

Supplemental progeny were collected as part of the upper Columbia River white sturgeon aquaculture program. Spawning and incubation occurred at the Kootenay Trout Hatchery (Wardner, British Columbia; FFSBC, 2011). Eggs were incubated in MacDonald jars (1.2 L of fertilised eggs per jar) at a constant 15°C using heated groundwater. MacDonald jars were set up in flow through with flow rates of 5 L/min until the neurulation stage was reached (~48 h) and then at 15 L/min until hatch. Increased flow rates after neurulation ensured eggs were constantly moving and helped reduce incidences of microbial infection (R. Ek, Kootenay Trout hatchery, personal communication). Time to hatch was approximately 9 days, and hatched larvae flowed into aluminium troughs where they were pooled as maternal family groups ($n=4$) and equal numbers of larvae from each maternal family were used in the experiment. Larval numbers were estimated on the basis of neurolisation percentages from each maternal family group (mean ± 1 SD = 76.9 ± 28.8). An estimated 336 220 larvae were stocked for the experiment, split evenly between the control and the modified sites. Larvae were acclimated at the hatchery to ambient Columbia River temperatures at the stocking site (10°C) for a 7-h period. Approximately 16 800 larvae were placed in each of 20 transport bags. The 14-L bags contained 4 L of 11°C water and 10 L of oxygen and were placed in an insulated trailer for a 5-h drive to the study site. Upon arrival, bags were cooled an additional 1°C to match the 10°C receiving water.

Larval stocking

Larvae were released 1 day posthatch over both the control and modified sites. Larvae were released during the daylight hours to encourage hiding (negative phototactic behaviour; Conte *et al.*, 1988; Loew and Sillman, 1998) within the substrate at both locations. Larvae were released on the river bottom immediately over the substrate using a weighted flexible hose (7.6 cm in diameter) attached to the middle of a 2-m long diffuser made of the same material. Rectangular holes (50 cm) cut out along the diffuser bottom allowed larvae the opportunity to disperse immediately into the substrate interstices. The diffuser had ropes attached to the outer ends so that it could be effectively manoeuvred on the river bottom from the boat. A cylindrical tank (40 L) equipped with a ball valve at the tank outlet was attached to the end of the hose on the boat. Before release, the valve was shut, one bag of larvae was loaded into the tank and the valve was opened to allow larvae to flow down and out of the release device. Additional water was poured down the system to flush out any remaining larvae within the hose. Equal numbers of larvae were released at six different locations within the upstream third of both the modified and control sites. Locations were

spaced evenly (horizontally and longitudinally) within the upper third of each area (Figure 1).

At the time of stocking, Revelstoke Dam was not operational so that flow velocities did not affect the ability of the larvae to hide within the substrate. One hour after stocking, flow was increased as per the typical daily operating regime. Daily discharge from the dam was not manipulated for this study and followed normal operations for power generation over the remainder of the experiment.

Monitoring

All monitoring and associated methods were replicated across both the control and the modified locations. Monitoring was conducted using passive sampling drift nets, which have shown to be effective in collecting sturgeon larvae dispersing downstream from spawning areas (Auer and Baker, 2002; Howell and McLellan, 2007; Golder, 2008). Dispersal is defined in this study as passive drift that is volitional or nonvolitional. Drift nets were arranged in three staggered rows of two (six nets per site; Figure 1). Drift nets were anchored to the river bottom using two claw anchors linked together with chain and attached to 10 m of cable. A single low profile buoy was attached to each of the anchor and the end of the cable. This allowed the cable to be retrieved and the net pulled and deployed without dislodging the anchor. The buoy from the anchor served as a backup in the event the buoy from the drift net was lost. During deployment, drift nets were attached to the cable and lowered slowly to the river bottom using a rope attached to the top of the frame. Drift nets were deployed and retrieved in a consistent fashion. When the drift nets were pulled, the net was rinsed and the collection cup was switched immediately for a clean one, and the net was redeployed to reduce the amount of time the net was inactive. Finally, we deployed a drift net between the two sampling sites to determine if larvae were able to disperse from one site into the other. This net was positioned near the bottom of both sites (Figure 1).

Drift nets were deployed and then larvae were stocked at 0730 on 3 July 2010. Drift nets were fished continuously throughout the study and were pulled and reset at 4-h intervals for the first 24 h after stocking, then every 6 h until 48 h after stocking, and then twice daily for the remainder of the experiment. During the period when nets were checked twice per day, the checks occurred once at dawn and once at dusk to test for temporal differences in dispersal (i.e. night versus day). When each drift net was retrieved, the number of larvae captured was enumerated. We preserved 10 larvae per net ($n=60$ per site) in PreferTM. Larvae were digitally imaged using a stereo microscope and camera (Nikon SMZ 745t) and measured (Nikon NIS Elements D, version 3.1, Melville, NY) for total length (mm) and yolk sac length and height (mm) within two months of preservation. Yolk sac volume (YSV)

was calculated using the yolk sac length and height data in the formula derived by Blaxter and Hempel (1963).

Physical data collection

We recorded discharge (cm) and temperature (°C) throughout the study period. Discharge was recorded hourly from Revelstoke Dam. Temperature was measured hourly at the study site using data loggers (Onset StowAway TidbitsTM; $\pm 0.2^\circ\text{C}$ accuracy). To determine whether hydraulic conditions varied between sites, we used an Acoustic Doppler Current Profiler (ADCP model 1200 kHz; Teledyne RDI Instruments, Poway, CA) to measure mean water velocity (m/s) and discharge (m^3/s) at transects within each site (Figure 1). A total of 10 transects were conducted at right angles to the flow, and end points were marked using a handheld GPS and fixed shoreline markers to ensure transects remained consistent within both sites. Measurements were recorded in water >1.2 m in depth. The distance to the shore from the transects end points was measured using a range finder and ADCP software (WinRiver II, version 2.07; Teledyne RDI Instruments) was used to interpolate the missing data to complete the discharge measurement. Total discharge estimates obtained from the ADCP software were qualitatively compared between the two sites within and between transects. Further, a paired *t*-test was used to compare differences in mean water velocities between the control and the modified sites.

Analysis

All statistical analyses were performed using R (R Development Core Team 2011, <http://www.rproject.org>). Data were tested for normality and homogeneity of variances using a Shapiro–Wilk test and by examining residuals versus fitted values in R. We used a general linear model to examine the effects and interactions of four independent variables on the number of larvae recaptured. Fixed effects included the site of capture and the net location within each site. The time of capture was included as a random variable in the model. Interactions among site, net location and time of capture were also examined. Discharge from Revelstoke Dam was included as a covariate in the analysis. We calculated mean discharge over the time the drift nets were deployed until the time they were retrieved for each sampling session.

White sturgeon larvae from a population in the Kootenai River at a similar latitude as the present study area have been demonstrated to have a nocturnal dispersal pattern (Kynard *et al.*, 2010). To test when larvae in our study initiated dispersal, we compared the distributions of the proportion of larvae collected during night sampling versus daylight sampling over the entire study period using a Kolmogorov–Smirnov test.

Larval white sturgeon size at hatch is variable within and among family groups (Parsley *et al.*, 2011), which could lead to a selective advantage in a larvae's ability to hide or initiate

drift. Further, larvae that are required to expend increased energy to hide have been demonstrated to grow slower and to use their yolk sac at a faster rate compared with larvae that have the ability to hide in suitable substrate (Crossman, 2008; McAdam, 2011). We tested for statistical differences in mean total length and YSV of larval white sturgeon collected at each of the two sites over the duration of the study using a two-factor analysis of variance with time (days) and sampling site as fixed factors. We also examined the interaction of time and site. We used Tukey's *post hoc* multiple comparison test to assess pairwise differences in total length between sites over time.

RESULTS

Monitoring was conducted continuously from 3 July 2010 through 25 July 2010. Daily mean temperatures were low and exhibited minimal variation throughout the experiment (mean $\pm$ 1 SD = 9.2 ± 0.8 °C). Mean daily discharge (297 ± 139 cm) varied by operational needs and ranged from 0 to 1160 cm during the experiment (Figure 2). Mean water velocities over the control site (1.84 ± 0.22 m/s) did not differ significantly ($t_{(372)} = 1.97$, $p = 0.83$) from those over the modified site (1.83 ± 0.29) when measured during an intermediate generation flow (~ 880 cm).

We collected 4329 larvae over the course of the experiment, which represented approximately 1% of the total number of larvae released. Significantly more larvae ($F_{1,556} = 8.07$, $P = 0.004$; Figure 2) were collected over the control site ($n = 4081$) compared with the modified site ($n = 241$). The catch per unit effort at these respective sites was 1.2 and 0.1 larvae per hour of sampling effort. During the experiment, only seven larvae were collected in the drift net at the cross over site (Figure 1), six of which were collected in the first day after larval stocking. This demonstrated that dispersing larvae were captured from within the site where they initiated

dispersal. The interaction of site of collection and time was significant ($F_{25,556} = 3.31$, $P < 0.001$), with more larvae collected during the early stages of sampling at both sites (Figure 2). Drift net configuration was consistent within each site (Figure 1). Although larvae were stocked in the upper third of each site, larval collections did not differ significantly by net location (e.g. upstream compared with downstream) at either site ($F_{5,556} = 0.53$, $P = 0.75$). Despite being variable, discharge was a significant predictor of larval collection, with more larvae collected during sampling periods of higher flow. Finally, except the first day after release, white sturgeon larvae were collected in significantly higher proportions during nighttime hours (0.81 ± 0.31) compared with daylight hours (0.16 ± 0.26 ; $D = 0.778$, $P < 0.001$).

There was a significant difference in larval total length over time between sites ($F_{22,460} = 67.65$, $P < 0.001$; Figure 3). Larvae collected from the modified site were significantly larger (16.3 ± 2.1 cm; $F_{1,460} = 111.6$, $P < 0.001$) compared with those collected at the control site (13.8 ± 2.2 cm) over the course of the experiment. Larvae that dispersed from the modified site were smaller in total length early on and were larger at the end of the experiment compared with larvae that dispersed from the control site (Figure 3). We found no significant differences between sites in mean $\pm$ 1 SD YSVs of collected larvae ($F_{1,327} = 0.026$, $P = 0.87$). However, there was a significant effect of time ($F_{13,327} = 2.5$, $P = 0.003$) on YSV with YSVs decreasing with increasing time after release. For example, YSVs were 9.91 ± 2.38 mm³ the first day after release compared with 4.73 ± 1.69 mm³ on day 14.

DISCUSSION

Results from this study demonstrate that modifications to embedded substrates at known white sturgeon spawning locations can result in enhanced conditions required for hiding at the yolk sac larvae stage. Larval hiding has been

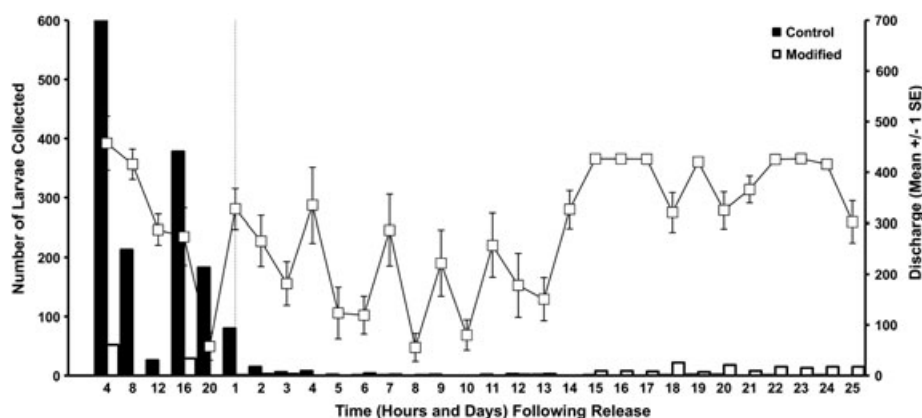


Figure 2. Number of larval white sturgeon collected at both control and modified sites by day with hourly discharge (cm) from Revelstoke Dam

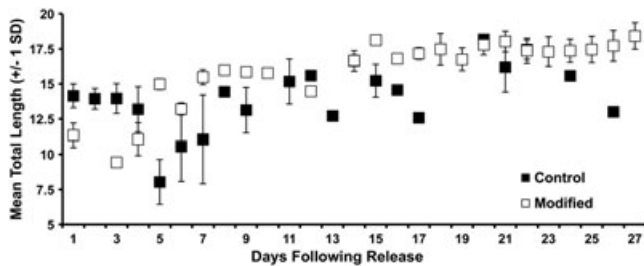


Figure 3. Total length (mm) of larval white sturgeon collected at both control and modified sites over the study period

identified as an important life history characteristic for white sturgeon (Kynard *et al.*, 2010; McAdam, 2011) and is thought to be critical for growth and development during early life history. The dispersal patterns observed indicated that larvae released at the enhanced site were better able to hide, to maintain their positions in the substrate, and to disperse volitionally as compared with larvae released over substrates in the control site. However, we did not find differences attributable to net location within either site, suggesting that some larvae were able to hide throughout both sites immediately after release. The ability for larvae to hide and put endogenous reserves towards growth and development rather than expend those reserves while searching for suitable hiding habitat or maintaining position in unsuitable habitat is important, as recruitment failure in the study area is thought to primarily occur during these early life stages (Gregory and Long, 2008). This study provides further support for the effectiveness of habitat mitigation focussed on substrate enhancements within known spawning grounds, as observed for other sturgeon species (Environment Illuminé Inc., 2010; Kerr *et al.*, 2010; Dumont *et al.*, 2011; Roseman *et al.*, 2011) and other fish species (Fuselier and Edds, 1995; Merz *et al.*, 2004; Pulg *et al.*, 2011).

The substrate material used in this study was selected on the basis of the best available information about white sturgeon spawning requirements and the ability to withstand high and variable discharges from Revelstoke Dam. Importantly, the design selected for the enhancement of sturgeon spawning areas will be highly dependent on the system, regulated or not. This is supported by the substantial variability in enhancement projects completed for other sturgeon species (Kerr *et al.*, 2010) as compared with other species with different ecologies that may have more consistent approaches across rivers (e.g. gravel augmentation for salmonids; Merz and Setka, 2004; Wheaton *et al.*, 2004; Harvey *et al.*, 2005; Brown and Pasternack, 2009). The material used in our study was a mix of primarily larger (90%; 200–300 mm diameter) angular stone with smaller rock (10%; 25–80 mm diameter) added to increase the quantity and quality of interstitial spaces. This material was considerably larger than those used for sturgeon spawning substrate in other field studies (e.g. Dumont *et al.*, 2011 20–30 cm; review by Kerr *et al.*, 2010) and that has been

suggested by experimental laboratory work (McAdam, 2011). Further, the area of the enhancement completed in our work was small (1000 m²) based on the relatively few adults that breed annually in the study area. In comparison, spawning enhancement projects completed for sturgeon in other rivers (reviewed by Kerr *et al.*, 2010) have ranged in size from 890 m² for lake sturgeon in the Eastmain River (Environment Illuminé Inc., 2010) to 8000 m² for lake sturgeon in the Des Prairies River (Dumont *et al.*, 2011), and to as large as 75 000 m² for stellate sturgeon (*Acipenser stellatus*) in the Kuban River (Chebanov *et al.*, 2008). Alternatively, other projects have focussed on building multiple smaller spawning sites in the same river (Folz and Meyers, 1985; Smith and Baker, 2005). However, most rivers where enhancement measures have been completed have had slower water velocities than those found in the present study area, which allows greater flexibility in design and placement options. System- and species-specific requirements need to be addressed when designing spawning habitat enhancement projects. The projects completed to date for sturgeon, including their variability, serve as important reference points that should be used in evaluating the feasibility of various options for future enhancement projects.

Substrate enhancement projects should be developed with the ecology of the species as the driving factor during the decision process. This can include metrics like the total number of breeding adults that might spawn annually or more specific requirements during early life stages (e.g. densities; physical attributes; Roseman *et al.*, 2011). Larger substrates dominated by boulders and cobbles provide incubating eggs and recently hatched larvae with interstitial space that promotes growth and development while at the same time excludes predators. Recent work by McAdam (2011) found that over porous substrates (1.2–15 cm diameter gravel to cobble), rapid interstitial hiding was observed. Further, hiding increased and predation by sculpins (*Cottus* spp) decreased in response to larger substrate size. Knowledge of interstitial space requirements about both construction and restoration of enhanced spawning habitat is critical during the design phase (Roseman *et al.*, 2011), especially for regulated rivers where sediment loads can be high and stream scouring flows might not be as frequent or effective as they were historically (Kerr *et al.*, 2010). Fish species in the study area (Ford and Thorley, 2011) include potential predators like bull trout (*Salvelinus confluentus*), mountain whitefish (*Prosopium williamsoni*) and prickly sculpin (*Cottus asper*), a species known to consume larval white sturgeon in differing habitat conditions (Gadomski and Parsley, 2005). These predators could have experienced different levels of accessibility or foraging effectiveness within the control compared with the enhanced area. Increased mortality due to predation could be another reason larval observations decreased over time. The habitat modifications may also have had both direct and indirect effects on other organisms in the

system. The work was completed the fall before our experiment, allowing possible establishment of other fish species and the macroinvertebrate community. Merz and Setka (2004) found that colonisation by macroinvertebrates occurred rapidly after the completion of gravel augmentation for salmon. This could be a positive benefit for fish in a regulated system where suitable benthic habitat may be lacking. Further, improved interstitial space could allow predators to remain in the study area during periods of peak flow compared with previous habitat, which had comparably less refuge. If predation is expected to be a significant source of mortality, then evaluation of fish ecologies in the project area is warranted during the design phase.

We found that larval sturgeon hiding at the modified site initiated dispersal primarily at night and at a consistent size of 17.5 mm total length (Figure 3). This suggests an important threshold that could be either developmental or size based. Increased variability in the size of larvae that dispersed from the control site suggested a reduced ability to hide effectively, irrespective of larval size. The size of larvae collected downstream of the modified site increased over time. We did not developmentally stage larvae between sites over time as our assumption was that development would be consistent over time and would not be predictive based on total length. Consistent development irrespective of larval growth has been demonstrated with shortnose sturgeon (*Acipenser brevirostrum*) larvae (E. Parker, personal communication). Most larval captures occurred shortly after release and later on when larvae initiated drift (Figure 2). We expected that nonvolitional dispersal would have been observed at the control site and would have been related to periods of higher flows. However, dispersal was better explained by substrate conditions (control versus modified) and the time of day. Revelstoke Dam is a peaking facility, but spawning occurs 6 km downstream, which may help attenuate changes in discharge. Larval white sturgeon size at dispersal downstream of a known spawning site on the lower Columbia River in Canada is at younger stage and size (mean $\pm$ 1 SD = 13.1 $\pm$ 1.2 mm; J. Crossman, unpublished data). Comparatively, Roseman *et al.* (2011) noted lake sturgeon larval dispersal from enhanced spawning habitat at 17–20 mm. Data about YSVs were only available before total absorption of the yolk, which occurred by day 15 after release. Furthermore, limited numbers of larvae from the modified site, due to their increased ability to hide after release, combined with ruptured yolk sacs due to the sampling gear prevented comparisons between sites over time. Results from this work contribute to knowledge at the larval stage for white sturgeon, a species where early life history is relatively understudied compared with other sturgeon species.

We found that substrate modifications improved habitat conditions for white sturgeon larvae within the first year after construction. Although the materials used were designed to withstand the flows anticipated in the study area with the

addition of a fifth-generation unit at Revelstoke Dam, uncertainty remains regarding the effectiveness of this habitat beyond the first few years. River regulation will eventually lead to habitat change, which may result in decreased substrate suitability over time. Visual assessments were conducted before the experiment (9 months after construction) and were qualitatively compared with assessments completed 1.5 years after construction. At 9 months after construction, the modified substrate was still within its original dimensions and appeared unchanged. At 1.5 years after construction (i.e. after a complete year of exposure to flow regulation), visual assessments indicated a notable change, with the modified substrates being widely dispersed downstream of the original location. Pulg *et al.* (2011) found that while both gravel addition and gravel cleaning created suitable spawning grounds for brown trout, these conditions were only maintained over the first two years, with reduced egg survival afterwards. Similarly, other studies have noted that improved spawning conditions only lasted 24 months (Merz and Setka, 2004) and suggest that mitigation must account for subsequent erosion or sedimentation (Kerr *et al.*, 2010). Smith and Baker (2005) found that constructed cobble spawning beds for lake sturgeon became filled with sand within the first decade. Other work found that sedimentation accumulating on the top layer of the substrate was dispersed by feeding activities of suckers and spawning activity of sturgeon (Environment Illimité Inc., 2010). The site constructed for this study is accessible at low or zero discharge from the Dam when reservoir conditions are low. This accessibility could be important for long-term maintenance but may not be possible in many rivers. Ultimately, monitoring the change in substrate conditions and the associated effect on suitability over the long term is critical to understanding if the habitat mitigation remains effective.

We demonstrated that in our study area, substrate modifications increased the suitability of hiding habitat for larval white sturgeon and resulted in an overall habitat enhancement for this life stage. However, further work is needed to determine if adult white sturgeon will selectively spawn on the modified habitat and if positive effects on egg retention and survival rates are realised. Other successful sturgeon enhancement projects have evaluated all life history stages (adults, eggs, larvae and juveniles) and can attribute benefits of the enhancement to all stages (e.g. Dumont *et al.*, 2011; Roseman *et al.*, 2011). We ceased sampling efforts during the larval dispersal phase from the enhanced site (Figure 2) as the original experimental design was based on existing early life history knowledge for this species, suggesting that hiding would not occur greater than 30 days posthatch. However, there were many unknowns given that this area represents the coldest spawning temperatures for white sturgeon spawning and early life stage rearing. Recent work by Parsley *et al.* (2011) on this temperature regime found that larvae can resist starvation

in these cold temperatures for up to 60 days, providing a large buffer before or during dispersal. These authors found that the time to hatch in the cool water temperatures was significantly delayed compared with more typical incubation temperatures ($>14^{\circ}\text{C}$), starting at day 10 and lasting for up to 11 days. These results combined with findings from this study will help in future habitat enhancement projects in this area. Finally, results from this study will be important for future mitigative efforts for sturgeon in regulated rivers where changes to important habitats have occurred.

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